STAT2004-7004-2904 – Assignment 4 Solutions

Marking guideline: each ✔ is worth 1 mark, but tutors have the freedom to mark each question as a whole. Half-marks may be awarded.

Question 1 (Testing exponential rates) (8 marks)

(a) (*2 marks*) Under $H_0 : \lambda = \theta$, all 15 observations come from the same exponential distribution, so the likelihood is

$$\begin{aligned} L(\lambda) &= \lambda^8 e^{-\lambda \sum_{i=1}^{8} X_i} \; \lambda^7 e^{-\lambda \sum_{j=1}^{7} Y_j} \\ &= \lambda^{15} e^{-\lambda(\sum X_i + \sum Y_j)} \checkmark \end{aligned}$$

The log-likelihood and its derivative are

$$\begin{aligned} \log L(\lambda) &= 15 \log \lambda - \lambda \left(\sum X_i + \sum Y_j \right) \\ \frac{\partial \log L(\lambda)}{\partial \lambda} &= \frac{15}{\lambda} - \left(\sum X_i + \sum Y_j \right) \checkmark \end{aligned}$$

Setting the derivative to 0 (the second derivative $-15/\lambda^2 < 0$, so this is a maximum) gives

$$\hat{\lambda} = \hat{\theta} = \frac{15}{\sum X_i + \sum Y_j} \; .$$

(b) (*1 mark*) Under $H_1 : \lambda \neq \theta$, the likelihood factorises into separate pieces for λ and θ:

$$\begin{aligned} L(\lambda, \theta) &= \lambda^8 e^{-\lambda \sum X_i} \; \theta^7 e^{-\theta \sum Y_j} \\ \log L(\lambda, \theta) &= 8 \log \lambda - \lambda \sum X_i + 7 \log \theta - \theta \sum Y_j \\ \frac{\partial \log L(\lambda, \theta)}{\partial \lambda} &= \frac{8}{\lambda} - \sum X_i = 0 \implies \hat{\lambda} = \frac{8}{\sum X_i} \\ \frac{\partial \log L(\lambda, \theta)}{\partial \theta} &= \frac{7}{\theta} - \sum Y_j = 0 \implies \hat{\theta} = \frac{7}{\sum Y_j} \checkmark \end{aligned}$$

(c) (*3 marks*)

$$\begin{aligned} \Lambda &= \frac{\sup_{H_0} L(\lambda, \theta | \boldsymbol{X}, \boldsymbol{Y})}{\sup_{H_1} L(\lambda, \theta | \boldsymbol{X}, \boldsymbol{Y})} \\ &= \frac{\left(\frac{15}{\sum X_i + \sum Y_j} \right)^{15} \exp(-15)}{\left(\frac{8}{\sum X_i} \right)^8 \left(\frac{7}{\sum Y_j} \right)^7 \exp(-8-7)} \checkmark \\ &= \frac{15^{15}}{8^8 \, 7^7} \left(\frac{\sum X_i}{\sum X_i + \sum Y_j} \right)^8 \left(\frac{\sum Y_j}{\sum X_i + \sum Y_j} \right)^7 \\ &= \frac{15^{15}}{8^8 \, 7^7} \; T^8 (1-T)^7 \; , \quad \checkmark \end{aligned}$$

since $1-T = \sum Y_j/(\sum X_i + \sum Y_j)$. The function $g(T) = T^8(1-T)^7$ is 0 at $T=0$ and $T=1$, increases on $[0, 8/15]$ and decreases on $[8/15, 1]$ (check: $\frac{d}{dT}\log g = 8/T - 7/(1-T)$). Hence small values of Λ correspond to either small or large values of T, and the GLRT rejects H_0 when $\{T \le c_1 \text{ or } T \ge c_2\}$. ✔

(d) (*1 mark*) Under H_0, $T \sim \text{Beta}(8, 7)$. We need $c_1 < c_2$ such that

$$0.01 = P(T \le c_1) + P(T \ge c_2) \ .$$

The simplest choice is to split the 1% equally between the two tails, i.e. $P(T \le c_1) = 0.005$ and $P(T \le c_2) = 0.995$:

```
> qbeta(0.005, 8, 7)
[1] 0.2234259
> qbeta(0.995, 8, 7)
[1] 0.8276016
```

So the decision rule is: reject H_0 if

$$\{T \le 0.223 \text{ or } T \ge 0.828\}.$$ ✔

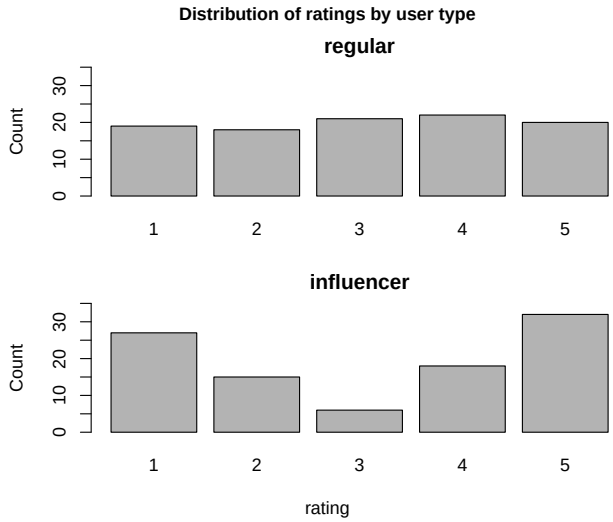
(Students may instead choose c_1, c_2 to also satisfy $c_1^8(1-c_1)^7 = c_2^8(1-c_2)^7$, so that the rule is exactly the GLRT; this gives $c_1 = 0.222$ and $c_2 = 0.827$. Either is fine.)

(e) (*1 mark*) [**Audio question**]: No. ✔ For alternatives with $\lambda > \theta$ the X's tend to be smaller than the Y's, so the most powerful test (by the Neyman–Pearson lemma) rejects only for small T; for $\lambda < \theta$ it rejects only for large T. These two one-sided tests are different, and the two-sided test from (c)–(d) splits its 1% between both tails, so it is less powerful than the one-sided test on each side. No single test is most powerful against every alternative $\lambda \ne \theta$, so no UMP test exists here.

Question 2 (Comparing ratings across groups) (8 marks)

0 marks for the whole question if students compute sample means, sample deviations, and use a two-sample t-test. The responses here are *categorical* not numerical, so it is meaningless to compute means and standard deviations.

(a) (*1 mark*) The ratings here are categorical. We should plot the counts (or proportions) in each category for each user type. This is perhaps clearest with a comparative bar chart: ✔



	rating				
user	1	2	3	4	5
regular	19	18	21	22	20
influencer	27	15	6	18	32

We immediately see that the opinions of "regular" users are fairly uniform across all five opinions whereas "influencers" are much more polarised on this issue.

(b) (*3 marks*) The null hypothesis is that the distribution of opinion ratings are the same across both user types ("regular" and "influencer"). The alternative hypothesis is that the distribution of opinion ratings are not necessarily the same across these two types of users. ✔ Mathematically, the null hypothesis can be written as

$$H_0 : \boldsymbol{p}_{\text{regular}} = \boldsymbol{p}_{\text{influencer}} \ ,$$

where the two vectors denote the underlying proportions of (strongly disagree, disagree, neutral, agree, and strongly agree) for each of the "regular" and "influencer" groups, respectively. The alternative hypothesis can then be written as $H_1 : \boldsymbol{p}_{\text{regular}} \neq \boldsymbol{p}_{\text{influencer}}$.

This can be considered a test of homogeneity. Calculating the expected values, either by hand or using software, gives the following expected counts (in parentheses) under the null:

	rating				
user	1	2	3	4	5
regular	19 (23.23)	18 (16.67)	21 (13.64)	22 (20.20)	20 (26.26)
influencer	27 (22.77)	15 (16.33)	6 (13.36)	18 (19.80)	32 (25.74)

To quantify the overall discrepancy between the observed and expected counts, we can use Pearson's χ^2 statistic:

$$\begin{aligned}\chi^2 &= \sum_{\text{cells}} \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} \\ &= \frac{(19-23.23)^2}{23.23} + \frac{(18-16.67)^2}{16.67} + \cdots + \frac{(32-25.74)^2}{25.74} \\ &= 13.15 \quad \text{on } (2-1)(5-1) = 4 \text{ degrees-of-freedom.} \checkmark\end{aligned}$$

The subsequent p-value for this test is $P(\chi^2_4 \geq 13.15) \approx 1.1\%$. There is moderate-to-strong evidence that the opinion ratings of "regular" users and "influencers" are different on this issue. ✔

(c) (*2 marks*) All expected and observed counts exceed 5 so the χ^2 approximation is not a worry.✔ However, as with any online poll, it is dubious whether the respondents in this poll are an independent and unbiased sample. Respondents may forward the poll to their friends/followers, or even recommend their friends/followers to vote in a particular way, and so that would break the independence assumption. Also, users who bother to participate in an online poll usually have stronger opinions about the issue than those who do not respond. ✔

(d) (*2 marks*) [**Audio question**]: The responses are categories ✔ which have been assigned arbitrary numerical values. So it is meaningless to compute means and standard deviations here ✔ . Moreover, the t-test would yield different results if we changed the arbitrary numerical values to -3, -1, 0, 1, 3, say, or any other set of 5 values.

Question 3 (Tuberculosis and blood type) (14 marks)

(a) (*1 mark*) In the context of this study, the null hypothesis is that tuberculosis severity is independent of, or unrelated to, blood type. The alternative hypothesis is that tuberculosis severity is related to blood type. ✔

(b) (*1 mark*) The likelihood function for $\boldsymbol{p}$ given the observed counts $\boldsymbol{x}$ is

$$\begin{aligned} L(\boldsymbol{p}|\boldsymbol{x}) &= \binom{n}{\boldsymbol{x}} \prod_{ij} (p_{ij})^{x_{ij}} \quad ✔ \\ &\propto (p_{11})^7 (p_{12})^5 (p_{13})^3 (p_{14})^{13} (p_{21})^{27} (p_{22})^{32} (p_{23})^8 (p_{24})^{18} (p_{31})^{55} (p_{32})^{50} (p_{33})^7 (p_{34})^{24} \ , \end{aligned}$$

where we have written out the product explicitly because there are "only" 12 terms in total. Of course, for bigger tables you would have to be comfortable with using the double product/double subscript notation.

(c) (*3 marks*) Substituting $p_{ij} = p_{i\bullet} \times p_{\bullet j}$ into the likelihood function gives

$$\begin{aligned} L &\propto (p_{1\bullet})^{7+5+3+13} \times (p_{2\bullet})^{27+32+8+18} \times (1 - p_{1\bullet} - p_{2\bullet})^{55+50+7+24} \\ &\quad \times (p_{\bullet 1})^{7+27+55} \times (p_{\bullet 2})^{5+32+50} \times (p_{\bullet 3})^{3+8+7} \times (1 - p_{\bullet 1} - p_{\bullet 2} - p_{\bullet 3})^{13+18+24} \\ &= (p_{1\bullet})^{x_{1\bullet}} (p_{2\bullet})^{x_{2\bullet}} (1 - p_{1\bullet} - p_{2\bullet})^{x_{3\bullet}} (p_{\bullet 1})^{x_{\bullet 1}} (p_{\bullet 2})^{x_{\bullet 2}} (p_{\bullet 3})^{x_{\bullet 3}} (1 - p_{\bullet 1} - p_{\bullet 2} - p_{\bullet 3})^{x_{\bullet 4}} \ , ✔ \end{aligned}$$

where we have written $p_{3\bullet} = 1 - p_{1\bullet} - p_{2\bullet}$, $p_{\bullet 4} = 1 - p_{\bullet 1} - p_{\bullet 2} - p_{\bullet 3}$, and $x_{i\bullet}$ for the total number of tuberculosis cases of severity i, and $x_{\bullet j}$ for the total number of cases of blood type j.

We can directly maximise the likelihood function, but as usual, it is easier to maximise the log-likelihood instead, which is given by

$$\begin{aligned} l &= x_{1\bullet} \log(p_{1\bullet}) + x_{2\bullet} \log(p_{2\bullet}) + x_{3\bullet} \log(1 - p_{1\bullet} - p_{2\bullet}) \\ &\quad + x_{\bullet 1} \log(p_{\bullet 1}) + x_{\bullet 2} \log(p_{\bullet 2}) + x_{\bullet 3} \log(p_{\bullet 3}) + x_{\bullet 4} \log(1 - p_{\bullet 1} - p_{\bullet 2} - p_{\bullet 3}) \end{aligned}$$

This function is smooth and concave in all the parameters, and so we can find the maximizer of the log-likelihood function by taking derivatives with respect to each parameter and setting them to 0. For $p_{1\bullet}$ and $p_{2\bullet}$, these equations are

$$\begin{aligned} 0 = \frac{\partial l}{\partial p_{1\bullet}} = \frac{x_{1\bullet}}{p_{1\bullet}} - \frac{x_{3\bullet}}{1 - p_{1\bullet} - p_{2\bullet}} &\iff \frac{x_{1\bullet}}{p_{1\bullet}} = \frac{x_{3\bullet}}{p_{3\bullet}}; \quad \text{and} \\ 0 = \frac{\partial l}{\partial p_{2\bullet}} = \frac{x_{2\bullet}}{p_{2\bullet}} - \frac{x_{3\bullet}}{1 - p_{1\bullet} - p_{2\bullet}} &\iff \frac{x_{2\bullet}}{p_{2\bullet}} = \frac{x_{3\bullet}}{p_{3\bullet}} \quad ✔ \end{aligned}$$

Hence, the MLEs for $p_{1\bullet}$, $p_{2\bullet}$ and $p_{3\bullet} = 1 - p_{1\bullet} - p_{2\bullet}$ must satisfy

$$\frac{x_{1\bullet}}{p_{1\bullet}} = \frac{x_{2\bullet}}{p_{2\bullet}} = \frac{x_{3\bullet}}{p_{3\bullet}} = k \ ,$$

for some common constant k. Summing $p_{i\bullet} = x_{i\bullet}/k$ over i gives $1 = n/k$, so $k = n$ and the solution is

$$\begin{aligned} \hat{p}_{1\bullet} &= x_{1\bullet}/n &&= 28/249 \\ \hat{p}_{2\bullet} &= x_{2\bullet}/n &&= 85/249 \\ \hat{p}_{3\bullet} &= x_{3\bullet}/n &&= 136/249 \quad ✔ \end{aligned}$$

By analogous calculations, we can show that

$$\begin{aligned}
\hat{p}_{\bullet 1} &= x_{\bullet 1}/n &&= 89/249 \\
\hat{p}_{\bullet 2} &= x_{\bullet 2}/n &&= 87/249 \\
\hat{p}_{\bullet 3} &= x_{\bullet 3}/n &&= 18/249 \\
\hat{p}_{\bullet 4} &= x_{\bullet 4}/n &&= 55/249
\end{aligned}$$

(d) (*1 mark*) Using the results from part (c), the expected proportion in cell (1,1), say, is $28/249 \times 89/249 = 0.0402$ and so the expected count would be $249 \times 0.0402 = 10.01$. In general, the expected count is $n\hat{p}_{i\bullet}\hat{p}_{\bullet j} = x_{i\bullet}x_{\bullet j}/n$. Similar calculations for all the other cells give us the following table of expected values (in parentheses):

	blood type				
tuberculosis severity	O	A	AB	B	
moderate or advanced	7 (10.01)	5 (9.78)	3 (2.02)	13 (6.18)	✔
minimal	27 (30.38)	32 (29.70)	8 (6.14)	18 (18.78)	
not present	55 (48.61)	50 (47.52)	7 (9.83)	24 (30.04)	

(e) (*1 mark*) The MLEs $\hat{p}_{ij}$ of each cell proportion p_{ij} under the alternative are simply the sample proportions in each cell, i.e. $\hat{p}_{ij} = x_{ij}/n$. ✔ For the first cell this is $\hat{p}_{11} = 7/249$, and so on for all the other cells in the table.

(f) (*2 marks*) You can directly plug in the estimated p_{ij}s under both the null and alternatives into the likelihood function. Alternatively, we can compute $-2\log\Lambda$ first via

$$\begin{aligned}
-2\log\Lambda &= 2\sum_{\text{cells}} \text{Observed} \times \log(\text{Observed}/\text{Expected}) \\
&= 2\Big[7\log(7/10.01) + 5\log(5/9.78) + \cdots + 24\log(24/30.04)\Big] \\
&= 14.21\ , \quad ✔
\end{aligned}$$

and so the GLRT statistic is $\Lambda = \exp(-14.21/2) = 0.00082 \approx 1/1219.5$ ✔ . In other words, the observed counts would have been 1219.5 times more likely to have been observed under the alternative than under the null.

(g) (*1 mark*) Using the results from part (d) Pearson's χ^2 statistic is computed as

$$\begin{aligned}
\chi^2 &= \sum_{\text{cells}} \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} \\
&= \frac{(7-10.01)^2}{10.01} + \frac{(5-9.78)^2}{9.78} + \cdots + \frac{(24-30.04)^2}{30.04} \\
&= 15.37 \quad ✔
\end{aligned}$$

Pearson's χ^2 statistic is numerically similar to the $-2\log\Lambda = 14.21$ statistic from part (f), and certainly conclusions using either statistic would be consistent with each other.

(h) (*2 marks*) The degrees-of-freedom is the number of free parameters under the alternative $(12 - 1 = 11)$ minus the number under the null $(2 + 3 = 5)$, i.e. $(3-1)(4-1) = 6$. The p-value obtained by comparing Pearson's χ^2-statistic to a χ^2 distribution with 6 degrees-of-freedom is $P(\chi^2_6 \geq 15.37) \approx 0.018 = 1.8\%$,✔ (using $-2\log\Lambda$ instead

gives $P(\chi^2_6 \geq 14.21) \approx 2.7\%$), so there is moderately strong evidence that tuberculosis severity is related to ABO blood type in this population. Looking at the table, Eskimos with blood type B appear to have more moderate/advanced tuberculosis than we would expect if severity were unrelated to blood type. ✔

(i) (*2 marks*) An alternative test is to use Fisher's exact test. We first need to argue that if tuberculosis is indeed unrelated to blood type, then

- the total number of Eskimos in the sample who happened to have blood type O, A, AB or B would have been exactly the same because this would just depend on the genetics of Eskimos and not on tuberculosis incidence ; and
- the total number of Eskimos in the sample who happened to have moderate/advanced, minimal, or no tuberculosis would have been exactly the same because each Eskimo would have contracted the disease (or not) due to some medical reason that is unrelated to their blood type.

Thus the margins of the table can be considered fixed.
Secondly, if the tuberculosis severity and blood type are indeed unrelated, then any table of counts with the same given margins would have been equally likely if the null hypothesis is true, allowing us to permute the entries in the table keeping the margins fixed. ✔ (for both these steps)

Running Fisher's exact test gives the following results: using 100,000 simulated tables,

```
table2 = matrix(c(7,5,3,13, 27,32,8,18, 55,50,7,24), nrow=3, byrow=TRUE)
chisq.test(table2, simulate.p.value=TRUE, B=100000)

Pearson's Chi-squared test with simulated p-value (based on 1e+05 replicates)
data:  table2
X-squared = 15.37, df = NA, p-value = 0.01802
```

The p-value is very similar at 1.8% and so we still conclude that there is moderately strong evidence that tuberculosis severity is related to ABO blood type. ✔ (Simulated p-values will vary slightly from run to run; anything around 1.7–1.9% is fine. `fisher.test(table2, simulate.p.value=TRUE)` gives about 2.1% and is also acceptable.)

Question 4 (Weight gain in pigs) (10 marks)

(a) (*3 marks*) Under the two-way ANOVA model with interactions, each observation has distribution

$$Y_{jki} \overset{\text{ind}}{\sim} N\Big(\mu + \alpha_j + \beta_k + \delta_{jk},\ \sigma^2\Big)\ , \quad j = 1, 2,\ k = 1, 2,\ i = 1, 2, 3\ .$$

The likelihood function is therefore

$$L(\mu, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\delta} | \boldsymbol{Y}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left\{-\frac{1}{2\sigma^2}\sum_{jki}(Y_{jki} - \mu - \alpha_j - \beta_k - \delta_{jk})^2\right\} \quad ✔$$

where $N = 2 \times 2 \times 3 = 12$ is the total sample size.

Maximising the likelihood in the mean parameters is equivalent to minimising the sum-of-squares term in the exponent, $①= \sum_{jki}(Y_{jki} - \mu - \alpha_j - \beta_k - \delta_{jk})^2$. This function ① is a quadratic in each mean parameter, and so taking partial derivatives and setting to zero will give the minimizer. Under the contrast constraints, the only free mean parameters are μ, α_2, β_2 and δ_{22}.

Provide a half-mark for each correct MLE below if it has been derived properly. No marks for only writing down the MLE with a faulty derivation, because they are provided in the question.

Start by taking the derivative with respect to δ_{22}:

$$\begin{aligned} 0 = \frac{\partial ①}{\partial \delta_{22}} &= -2\sum_i (Y_{22i} - \mu - \alpha_2 - \beta_2 - \delta_{22}) \\ 3\delta_{22} &= \sum_i (Y_{22i}) - 3\mu - 3\alpha_2 - 3\beta_2 \\ \delta_{22} &= \overline{Y}_{22\bullet} - \mu - \alpha_2 - \beta_2 \end{aligned}$$

Next, take the derivative with respect to α_2 (recalling $\alpha_1 = \beta_1 = \delta_{21} = 0$):

$$\begin{aligned} 0 = \frac{\partial ①}{\partial \alpha_2} &= -2\sum_{ki} (Y_{2ki} - \mu - \alpha_2 - \beta_k - \delta_{2k}) \\ 6\alpha_2 &= \sum_{ki} (Y_{2ki}) - 6\mu - 3\beta_2 - 3\delta_{22} \\ \alpha_2 &= \overline{Y}_{2\bullet\bullet} - \mu - \frac{1}{2}\beta_2 - \frac{1}{2}\delta_{22} \end{aligned}$$

Using the previous expression for δ_{22}, and $\overline{Y}_{2\bullet\bullet} = \frac{1}{2}(\overline{Y}_{21\bullet} + \overline{Y}_{22\bullet})$ (balanced design):

$$\begin{aligned} \alpha_2 &= \overline{Y}_{2\bullet\bullet} - \mu - \frac{1}{2}\beta_2 - \frac{1}{2}(\overline{Y}_{22\bullet} - \mu - \alpha_2 - \beta_2) \\ \frac{1}{2}\alpha_2 &= \overline{Y}_{2\bullet\bullet} - \frac{1}{2}\overline{Y}_{22\bullet} - \frac{1}{2}\mu \\ \frac{1}{2}\alpha_2 &= \frac{1}{2}\overline{Y}_{21\bullet} - \frac{1}{2}\mu \\ \alpha_2 &= \overline{Y}_{21\bullet} - \mu \end{aligned}$$

In the same manner (differentiating with respect to β_2) we obtain the following expression for β_2:

$$\beta_2 \;=\; \overline{Y}_{12\bullet} - \mu$$

Substituting the expressions for α_2 and β_2 into δ_{22} yields

$$\begin{aligned}\delta_{22} &= \overline{Y}_{22\bullet} - \mu - (\overline{Y}_{21\bullet} - \mu) - (\overline{Y}_{12\bullet} - \mu)\\ \delta_{22} &= \overline{Y}_{22\bullet} - \overline{Y}_{21\bullet} - \overline{Y}_{12\bullet} + \mu\end{aligned}$$

Now differentiate with respect to μ

$$\begin{aligned}0 = \frac{\partial \textcircled{1}}{\partial \mu} &= -2\sum_{jki}(Y_{jki} - \mu - \alpha_j - \beta_k - \delta_{jk})\\ 12\mu &= \sum_{jki}(Y_{jki}) - 6\alpha_2 - 6\beta_2 - 3\delta_{22}\\ \mu &= \overline{Y}_{\bullet\bullet\bullet} - \frac{1}{2}\alpha_2 - \frac{1}{2}\beta_2 - \frac{1}{4}\delta_{22}\end{aligned}$$

Upon substituting all previously obtained expressions we obtain:

$$\begin{aligned}\mu &= \overline{Y}_{\bullet\bullet\bullet} - \frac{1}{2}(\overline{Y}_{21\bullet} - \mu) - \frac{1}{2}(\overline{Y}_{12\bullet} - \mu) - \frac{1}{4}(\overline{Y}_{22\bullet} - \overline{Y}_{21\bullet} - \overline{Y}_{12\bullet} + \mu)\\ \frac{1}{4}\mu &= \overline{Y}_{\bullet\bullet\bullet} - \frac{1}{4}\left(\overline{Y}_{21\bullet} + \overline{Y}_{12\bullet} + \overline{Y}_{22\bullet}\right)\\ \mu &= 4\overline{Y}_{\bullet\bullet\bullet} - \overline{Y}_{21\bullet} - \overline{Y}_{12\bullet} - \overline{Y}_{22\bullet} \;=\; \overline{Y}_{11\bullet}\;,\end{aligned}$$

since $4\overline{Y}_{\bullet\bullet\bullet} = \overline{Y}_{11\bullet} + \overline{Y}_{12\bullet} + \overline{Y}_{21\bullet} + \overline{Y}_{22\bullet}$ in a balanced design.
So the MLEs are

$$\begin{aligned}\hat{\mu} &= \overline{Y}_{11\bullet}\\ \hat{\alpha}_2 &= \overline{Y}_{21\bullet} - \overline{Y}_{11\bullet}\\ \hat{\beta}_2 &= \overline{Y}_{12\bullet} - \overline{Y}_{11\bullet}\\ \hat{\delta}_{22} &= \overline{Y}_{22\bullet} - \overline{Y}_{21\bullet} - \overline{Y}_{12\bullet} + \overline{Y}_{11\bullet}\end{aligned}$$

(A quicker alternative: the model has four free mean parameters for four group means, so the fitted group means must equal the sample group means, $\hat{\mu}+\hat{\alpha}_j+\hat{\beta}_k+\hat{\delta}_{jk} = \overline{Y}_{jk\bullet}$. Solving these four equations under the constraints gives the same MLEs. Full marks if this is argued properly.)

(b) (*2 marks*) Starting with the identity,

$$\begin{aligned}
\mathrm{SS}_{\mathrm{Total}} &= \sum_{jki}(Y_{jki} - \overline{Y}_{\bullet\bullet\bullet})^2 \\
&= \sum_{jki}\left[(Y_{jki} - \overline{Y}_{jk\bullet}) + (\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet}) + (\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet}) + (\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet})\right]^2 \\
&= \sum_{jki}\left[(Y_{jki} - \overline{Y}_{jk\bullet})^2 + (\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet})^2 + (\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet})^2 + (\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet})^2\right] \\
&+ 2\sum_{jki}(Y_{jki} - \overline{Y}_{jk\bullet})(\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet}) + 2\sum_{jki}(Y_{jki} - \overline{Y}_{jk\bullet})(\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet}) \\
&+ 2\sum_{jki}(\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet})(\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet}) + 2\sum_{jki}(Y_{jki} - \overline{Y}_{jk\bullet})(\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet}) \\
&+ 2\sum_{jki}(\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet})(\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet}) \\
&+ 2\sum_{jki}(\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet})(\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet})
\end{aligned}$$

Must keep each of the cross-terms ✔ and show/explain all cross-terms contain deviations away from mean, so yield 0 ✔

Each cross-term is zero because, in this balanced design:

- the three cross-terms involving $(Y_{jki} - \overline{Y}_{jk\bullet})$ multiply it by something that does not depend on i, and $\sum_i(Y_{jki} - \overline{Y}_{jk\bullet}) = 0$ for every j, k;
- $\sum_{jki}(\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet})(\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet}) = 3\sum_j(\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet})\sum_k(\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet}) = 0$;
- $\sum_k(\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet}) = 2\overline{Y}_{j\bullet\bullet} - 2\overline{Y}_{j\bullet\bullet} - 2\overline{Y}_{\bullet\bullet\bullet} + 2\overline{Y}_{\bullet\bullet\bullet} = 0$ for every j, which kills the second-last cross-term; similarly summing over j kills the last one.

Hence

$$\begin{aligned}
\mathrm{SS}_{\mathrm{Total}} &= \sum_{jki}(Y_{jki} - \overline{Y}_{jk\bullet})^2 + \sum_{jki}(\overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet\bullet\bullet})^2 \\
&+ \sum_{jki}(\overline{Y}_{\bullet k\bullet} - \overline{Y}_{\bullet\bullet\bullet})^2 + \sum_{jki}(\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet})^2 \\
&= \mathrm{SS}_{\mathrm{resid}} + \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{AB}
\end{aligned}$$

(c) (*1 mark*) We know that the residual sum-of-squares $\sum_i(Y_{jki} - \overline{Y}_{jk\bullet})^2$ within each group (j, k) has a $\sigma^2\chi^2_2$ distribution with $r - 1 = 2$ degrees-of-freedom. ✔ Because all $2 \times 2 = 4$ groups are independent of each other, the overall residual sum-of-squares $\mathrm{SS}_{\mathrm{resid}} = \sum_{jki}(Y_{jki} - \overline{Y}_{jk\bullet})^2$ obtained by summing over the four groups therefore has a $\sigma^2\chi^2_8$ distribution.

(d) (*1 mark*) We know that the residual sum-of-squares $\sum_i(Y_{jki} - \overline{Y}_{jk\bullet})^2$ within each group (j, k) is independent of the sample mean $\overline{Y}_{jk\bullet}$ of that group. Hence, the overall residual sum-of-squares $\mathrm{SS}_{\mathrm{resid}} = \sum_{jki}(Y_{jki} - \overline{Y}_{jk\bullet})^2$ is independent of all the group means $\{\overline{Y}_{jk\bullet}\}$ ✔ . Note that the interaction sum-of-squares $\mathrm{SS}_{AB} = \sum_{jki}(\overline{Y}_{jk\bullet} - \overline{Y}_{j\bullet\bullet} - \overline{Y}_{\bullet k\bullet} + \overline{Y}_{\bullet\bullet\bullet})^2$ depends only on (functions of) the group means $\{\overline{Y}_{jk\bullet}\}$. Hence, $\mathrm{SS}_{\mathrm{resid}}$ is independent of SS_{AB}.

(e) (*1 mark*) Using part (c) and the given result, the F-ratio is

$$F = \frac{\text{MS}_{AB}}{\text{MS}_{\text{residual}}} := \frac{\text{SS}_{AB}/df_{AB}}{\text{SS}_{\text{residual}}/df_{\text{residual}}} \sim \frac{\chi^2_{df_{AB}}/df_{AB}}{\chi^2_{df_{\text{resid}}}/df_{\text{resid}}} \checkmark$$

(the σ^2's cancel), where the numerator is independent of the denominator by result (d). This is the definition of an F distribution with numerator degrees-of-freedom df_{AB} and denominator degrees-of-freedom df_{residual}.

(f) (*2 marks*) [Labelled as a second "(e)" on the assignment sheet.] Completing the two-way ANOVA table gives the following table:

Source	df	SS	MS	F	P
VitaminB12	1	218700	218700	60.33	< 0.0001
Antibiotics	1	19200	19200	5.30	≈ 0.05
VitaminB12:Antibiotics	1	172800	172800	47.67	0.0001
Residuals	8	29000	3625		✔
Total	11	439700			

Entries in red are the ones students need to fill in. Here $\text{SS}_{AB} = 439700 - 218700 - 19200 - 29000 = 172800$ by part (b), $df_{\text{resid}} = 11 - 1 - 1 - 1 = 8$, $\text{MS} = \text{SS}/df$, and each F is its MS divided by $\text{MS}_{\text{resid}} = 3625$, e.g. $F = 172800/3625 = 47.67$ for the interaction, with p-values from the $F_{1,8}$ distribution.

Note to markers: the y-axis of the plot on the assignment sheet is labelled "kg" but the text says weight gain is in grams. Do not penalise students for using either unit.

Looking at the test of interaction first, we find very strong evidence of an interaction effect - that is, there is very strong evidence that the effect of vitamin B12 depends on whether or not antibiotics are present, and vice versa, with an associated p-value of $P(F_{1,8} \geq 47.67) = 0.0001 = 0.01\%$ ✔ Looking at the graph of the data, we see that Vitamin B12 has a strong positive effect on weight gain in pigs when antibiotics are present but little effect when antibiotics are absent. We should not look at the main effects of each factor when interaction(s) are present.

Question 5 (STAT2904 only) (10 marks)

(a) (*4 marks*) The likelihood function is

$$L(\theta,\nu) = \begin{cases} \theta^n \nu^{n\theta} \prod_{i=1}^n \frac{1}{X_i^{\theta+1}} , & \text{if } \nu \le \text{all } X_i , \\ 0 , & \text{otherwise} \end{cases} \quad ✔$$

For each $\theta > 0$ the likelihood is a monotonically increasing function in ν and so the larger ν the higher the likelihood. However, ν cannot exceed any X_i, otherwise the likelihood switches to 0. Hence $\hat{\nu} = \min\{X_i\}$. ✔On the other hand, fixing $\hat{\nu} = \min\{X_i\}$, the likelihood is a smooth function in θ and the log-likelihood function is

$$l = n\log\theta + n\theta\log(\hat{\nu}) - (\theta+1)\sum_{i=1}^n \log(X_i)$$

Differentiating the log-likelihood with respect to θ and setting to 0 gives

$$0 = \frac{n}{\theta} + n\log(\hat{\nu}) - \sum_{i=1}^n \log(X_i) ✔$$

and so

$$\begin{aligned} \hat{\theta} &= \frac{n}{\sum_{i=1}^n \log(X_i) - n\log(\hat{\nu})} \\ &= \frac{n}{\sum_{i=1}^n \log(X_i) - n\log(\min\{X_i\})} \quad ✔ \\ &= \frac{n}{\sum_{i=1}^n \log(X_i/\min\{X_i\})} = \frac{n}{T(\mathbf{X})} \end{aligned}$$

(b) (*1 mark*) No, the MLE for ν is functionally independent of the value of θ and so knowing $\theta = 1$ does not change the MLE for ν. ✔

(c) (*5 marks*) The GLRT statistic is

$$\begin{aligned} \Lambda &= \frac{\sup_{\theta=1,\,\nu} L(\theta,\nu)}{\sup_{\theta,\,\nu} L(\theta,\nu)} = \frac{L(1,\hat{\nu})}{L(\hat{\theta},\hat{\nu})} ✔ \\ &= \frac{\hat{\nu}^n / \prod_i X_i^2}{\hat{\theta}^n \hat{\nu}^{n\hat{\theta}} / \prod_i X_i^{\hat{\theta}+1}} ✔ \\ &= \hat{\theta}^{-n} \left(\frac{\prod_i X_i}{\hat{\nu}^n} \right)^{\hat{\theta}-1} \\ &= \hat{\theta}^{-n} \exp\{T(\hat{\theta}-1)\} , \quad \text{since } T = \log\left(\prod_i X_i/\hat{\nu}^n \right) \\ &= (T/n)^n \exp(n-T) ✔ , \quad \text{using } \hat{\theta} = n/T, \end{aligned}$$

which is a function of the statistic T alone.

Differentiating, $\frac{d}{dT}\log\Lambda = \frac{n}{T} - 1$, which is positive for $T < n$ and negative for $T > n$. So Λ increases in T up to its maximum of 1 at $T = n$, then decreases ✔.
Hence, small values of Λ are equivalent to small or large values of T, i.e. the GLRT rejects H_0 when $\{T \le c_1 \text{ or } T \ge c_2\}$. ✔

STAT2904 Bonus Questions (5 marks)

Note: the assignment sheet writes $\prod_{i=1}^{22}$ and $(\min x_i)^{22}$ in part (d), but the sample size is $n = 25$. Accept either 22 or 25 as long as the student is consistent.

(d) (*2 marks*) The Pareto cdf is $F(x) = 1 - (\nu/x)^\theta$ for $x \geq \nu$, so by the inverse-cdf method $X = \nu\, U^{-1/\theta}$ with $U \sim U(0,1)$. ✔

```
set.seed(2026)
rpareto = function(n, theta, nu) nu * runif(n)^(-1/theta)
Tstat = function(x) sum(log(x)) - length(x)*log(min(x))
x = rpareto(25, theta=1, nu=2.1)
Tstat(x)
[1] 37.36862
```

The observed value will differ between students (it depends on the random seed). ✔

(e) (*1 mark*)

```
Ts = replicate(10000, Tstat(rpareto(25, theta=1, nu=2.1)))
```

✔

(f) (*1 mark*)

```
quantile(Ts, c(0.025, 0.975))
    2.5%    97.5%
15.33347 34.41351
```

So at the 5% level we reject $H_0 : \theta = 1$ if $T \leq 15.3$ or $T \geq 34.4$ (approximately). ✔

(g) (*1 mark*) Repeating with other values of ν:

ν	2.5%-tile	97.5%-tile
1.3	15.39	34.49
2.1	15.33	34.41
2.7	15.18	34.19
3.4	15.35	34.44

The cutoffs do not change (beyond simulation noise) as ν changes. ✔This is because $\log(X_i/\nu) \sim \text{Exp}(\theta)$, so $T = \sum_i \log(X_i/\nu) - n \log(\min_i X_i/\nu)$ does not depend on ν at all, so the test is free of the nuisance parameter. (In fact, under H_0, $T \sim \text{Gamma}(n-1, 1)$, with exact 2.5% and 97.5% quantiles 15.38 and 34.51 for $n = 25$. Students are not expected to show this.)

Question 6 (Bonus question for all students) [4 marks]

(a) (*2 marks*) Since $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \mu^2)$, the sample mean has distribution $\overline{X} \sim N(\mu, \mu^2/n)$. ✔Standardising,

$$Z = \frac{\overline{X} - \mu}{\mu/\sqrt{n}} = \sqrt{n}\left(\frac{\overline{X}}{\mu} - 1\right) \sim N(0,1) \ .$$

Z is a function of the data and μ only, and its distribution does not depend on μ, so it is a pivot. ✔

(b) (*2 marks*) Let $z = 1.645$ be the 95%-tile of $N(0,1)$. Then

$$0.90 = P\left(-z \le \sqrt{n}\left(\frac{\overline{X}}{\mu} - 1\right) \le z\right) = P\left(1 - \frac{z}{\sqrt{n}} \le \frac{\overline{X}}{\mu} \le 1 + \frac{z}{\sqrt{n}}\right) \ .✔$$

Since $\mu > 0$, inverting the inequalities gives the 90% confidence interval

$$\left[\ \frac{\overline{X}}{1 + 1.645/\sqrt{n}}\ ,\ \frac{\overline{X}}{1 - 1.645/\sqrt{n}}\ \right] \ ,✔$$

which is valid when $n \ge 3$ (so that $1 - 1.645/\sqrt{n} > 0$) and $\overline{X} > 0$. (If $n \le 2$ the upper limit is $+\infty$.) Note that the interval is not symmetric about $\overline{X}$, reflecting the larger variability for larger μ.

Other valid pivots are also acceptable, e.g. $(n-1)S^2/\mu^2 \sim \chi^2_{n-1}$, which gives the interval

$$\left[\ S\sqrt{\frac{n-1}{\chi^2_{n-1,0.95}}}\ ,\ S\sqrt{\frac{n-1}{\chi^2_{n-1,0.05}}}\ \right] \ .$$